Ch 6.2: Shrinkage (Part 2)

Lecture 15 - CMSE 381

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Wed, Mar 2, 2022

Announcements

Last time:

- Ridge Regression
- Lasso

This lecture:

- Optimization formulation of ridge/lasso
- Chooisng λ via cross-validation

Announcements:

- No homework this week
- Fri 3/4: Midterm
  - In person
  - Difficulty on par with homeworks
  - Cheat sheet: One 8.5x11 paper.
  - Handwritten
  - No photocopies

Section 1

Last time

Goal

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0. *Ridge regression*
- Better yet, make a bunch of them *actually* 0! *Lasso*

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

Find $\hat{\beta}$ to minimize....

Least Squares:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \underbrace{\sum_{j=1}^{p}\beta_j x_{ij}}_{\hat{y}_i}\right)^2$$

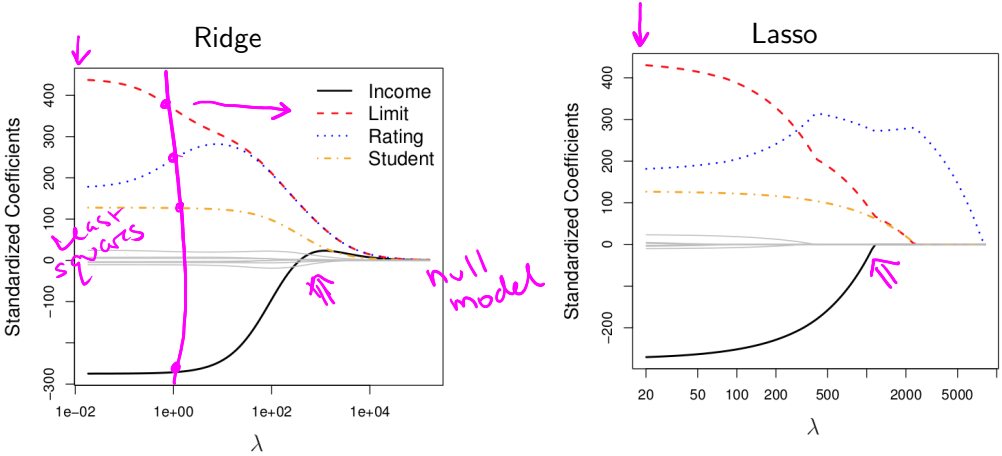
Ridge:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

The Lasso:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

Example on Credit data set

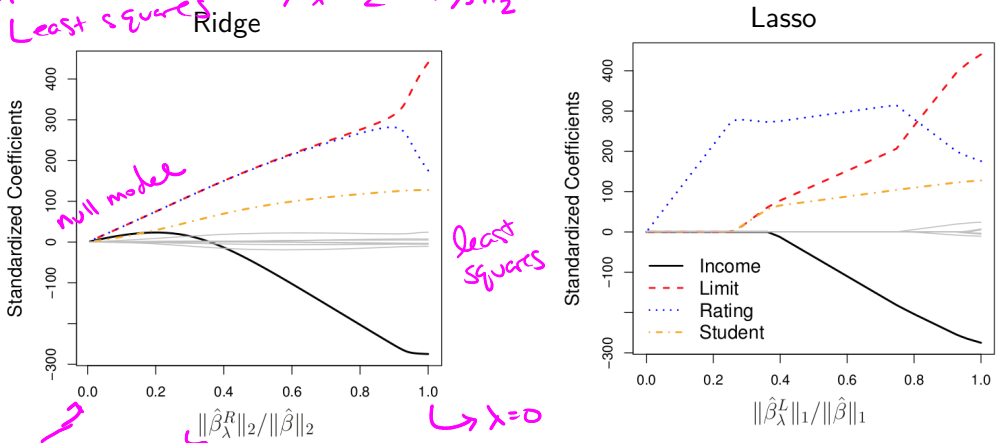


More example on Credit data set

$\lambda = 0$

Least squares

$\|\hat{\beta}^R_\lambda\|_2 = \|\hat{\beta}\|_2$



null model

least squares

$\lambda = 0$

$\lambda \to \infty$

$\|\hat{\beta}^R_\lambda\|_2 \Rightarrow 0$

$\|\hat{\beta}\|_2 = \left(\sum \beta_i^2\right)^{1/2}$

Scale equivavariance (or lack thereof)

Scale equivariant: Multiplying a variable by c just returns a coefficient multiplied by $1/c$

Least squares **is** scale equivariant.
Ridge/Lasso **are very much not**.

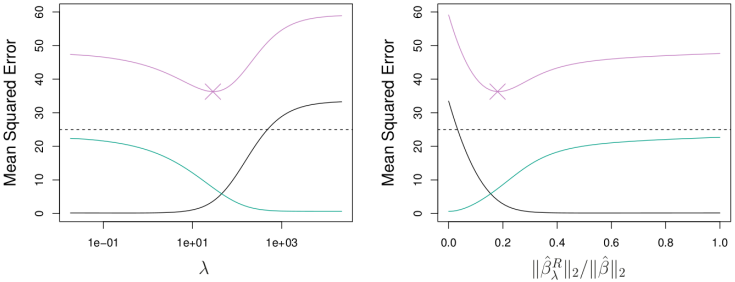
Solution: standardize predictors

$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

each data pt

divide by std dev of column (X_j variable)

Bias-Variance tradeoff



Squared bias (black), variance (green), and test mean squared error (purple) for simulated data.

Section 2

Optimization Formulation

Another formulation for Ridge Regression

Find β to minimize:

$$RSS + \lambda \underbrace{\sum_{j=1}^{p} \beta_j^2}_{\ell_2\text{-penalty}}$$

Find β to minimize

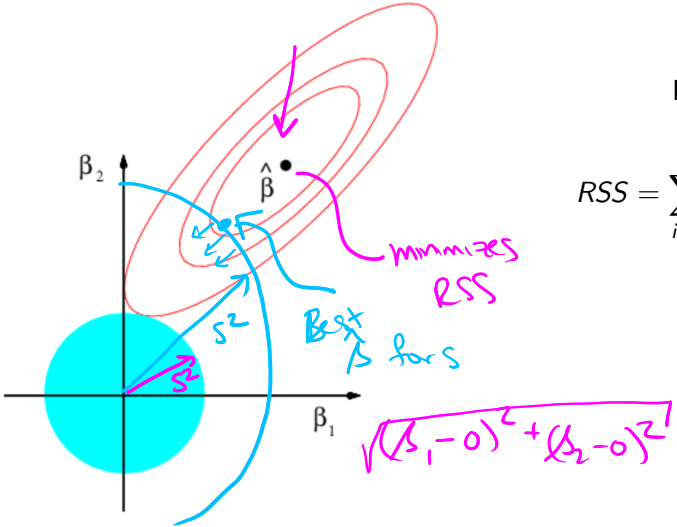
$$RSS$$

subject to

$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

related to but not quite the same as λ

Visualization using disks



Find β to minimize

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

subject to

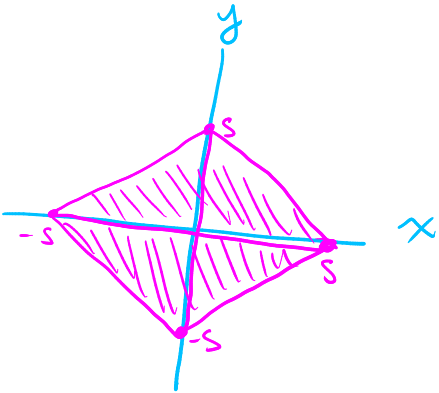
$$\sum_{j=1}^{p}\beta_j^2 \leq s$$

$$\sqrt{\sum \beta_i^2} \leq s^2$$

What about ℓ_1?

$$\|\beta\|_1 = \sum |\beta_i|$$

What does the set of points (x, y) for which $\|(x, y)\|_1 \leq s$ look like?

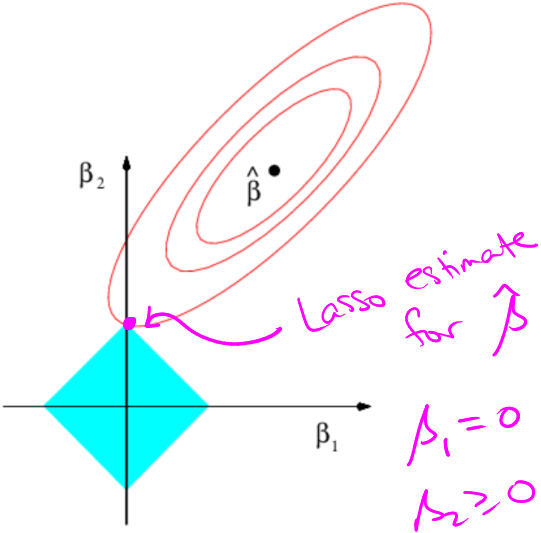


$$\underbrace{\|(0, y)\|}_{0+|y|} \leq s$$

$$0 + |y| \leq s$$

Same game for Lasso

Non smoothness of diamond shape $\Rightarrow$ minimizer on axes $\Rightarrow$ 0 coeff



Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to

$$\sum_{j=1}^{p} |\beta_j| \leq s$$

Same game for subset selection

Find β to minimize

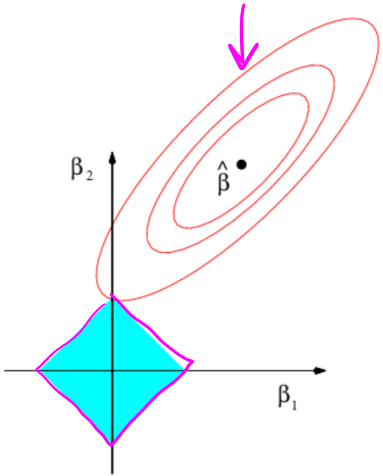
$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

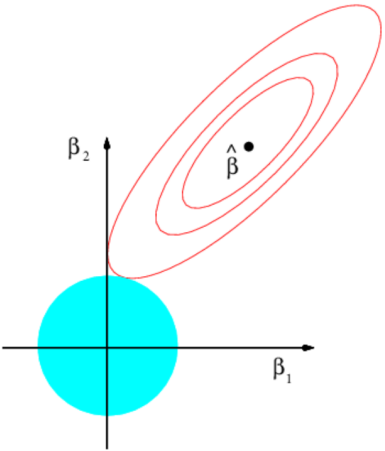
subject to

$$\sum_{j=1}^{p} \mathrm{I}(\beta_j \neq 0) \leq s$$

- Similar format but computationally gross
- Lasso/Ridge are computationally feasible ~~versions~~ approximations of subset

Using this visual to understand why lasso gets us zero values





Section 3

Tuning parameter selection

Find x to minimize

$g(x)$

subject to

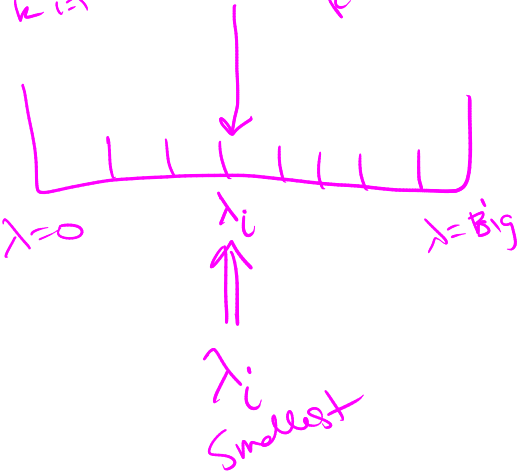
$f(x) \leq S$

Using Cross-Validation to find λ

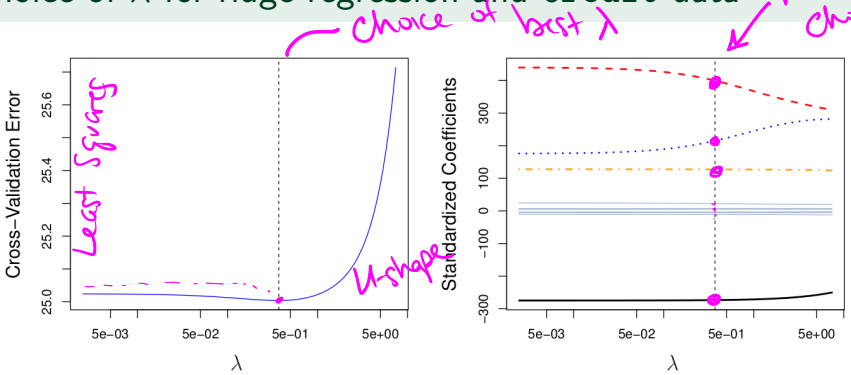
- Choose a grid of λ values
- Compute the (k-fold) cross-validation error for each value of λ
- Select the tuning parameter value λ for which the CV error is smallest.
- The model is re-fit using all of the available observations and the selected value of the tuning parameter.

$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} MSE_i$

fit model using $RSS + \lambda_i \sum \beta_j^2$

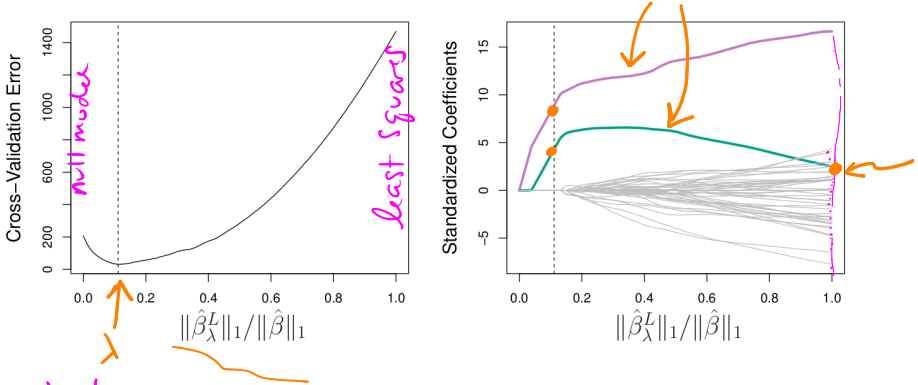


LOOCV choice of λ for ridge regression and `Credit` data



- Not very pronounced dip
- Probably just use least squares

10-fold CV choice of λ for lasso and simulated data



Simulated data

* $p = 45$ variables
* $n = 50$ data points

TL;DR

Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to:

Least Squares:
No constraints

Ridge:

$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

Inside circle

The Lasso:

$$\sum_{j=1}^{p} |\beta_j| \leq s$$

Inside diamond

Next time

Next time:

- Midterm!

This week:

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Bootstrap ⇝ std error of estimate

↳ resample data
get estimate
repeat

Validation / k-fold / LOOCV

Subsets of data
fit model on set A
Score on set B
↳ test error approx

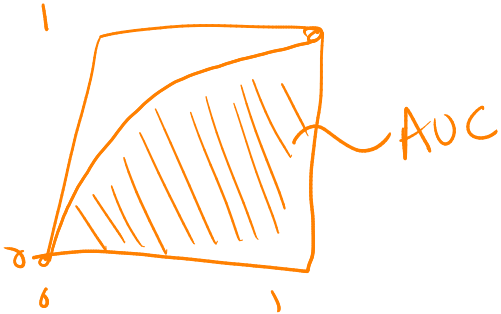
Section 4

Blank pages for notes

Prediction vs Inference

which variables affect the outcome most

See new data pt
predict what happens



Dummy variables: Input variable X_i qualitative

$X_i \in \{A, B, C, D\}$ (A: pikes, B: bananas, C: lemon tree, D: dragonfruit)

1 2 3 4

1 4 2 3

X_i	X_i^A	X_i^B	X_i^C	$\cancel{X_i^D}$
A	1	0	0	0
D	0	0	0	1
D	0	0	0	1
C	0	0	1	0
B	0	1	0	0